

Multi-Model Boundary Refinement for Person Segmentation

Artificial Intelligence Applications Final Project

Abstract

This project studies a practical person segmentation pipeline that improves boundary quality by combining existing models and lightweight image-processing modules. The proposed pipeline uses YOLO detection for person localization, Segment Anything Model (SAM) for promptable mask generation, Canny edge detection for boundary-based scoring, and GrabCut for foreground refinement. Instead of training a new neural network, the project compares multiple combinations of these modules on a selected 12-image test set. The best current version, YOLO-guided SAM with edge refinement and GrabCut, achieved the highest mean edge alignment score of 0.8855. Since no manually annotated ground-truth masks are available yet, the reported score should be interpreted as a preliminary boundary-alignment metric rather than final segmentation accuracy.

Keywords: person segmentation, SAM, YOLO, GrabCut, Canny edge detection, boundary refinement

1. Introduction

Person segmentation is used in tasks such as background removal, visual editing, surveillance preprocessing, and human-object analysis. A key challenge is not only detecting the person, but also producing a clean mask boundary that follows visible body edges.

The initial project direction used SAM automatic mask generation. Although SAM can produce strong masks, fully automatic mask selection may still be unstable when the image contains multiple people, complex backgrounds, or large foreground regions. The midterm feedback suggested adding more models and improving result precision. Therefore, the final project was extended into a multi-model comparison pipeline.

The main research question is whether a YOLO-guided SAM pipeline can improve boundary alignment compared with using SAM automatic masks or YOLO-Seg alone.

2. Related Work

2.1 Segment Anything Model

SAM is a promptable segmentation model that can generate masks from prompts such as points, boxes, and masks. Kirillov et al. introduced SAM as a general segmentation foundation model with strong zero-shot transfer ability. This project uses SAM as the main segmentation backbone because it can generate person masks from an automatically produced bounding-box prompt.

2.2 YOLO Object Detection

YOLO-based models are widely used for real-time object detection. A review by Terven and Cordova-Esparza describes the development of YOLO architectures through YOLOv8 and YOLO-NAS. In this project, YOLO is used as a localization module rather than the final segmentation model. The detected person bounding box is converted into a prompt for SAM.

2.3 Prompt Quality, GrabCut, and Edge Detection

SAM performance depends on prompt type and prompt quality. Studies by Gaus et al. and Mazurowski et al. support the idea that less ambiguous prompts, especially box prompts, can improve SAM mask quality. GrabCut is used as a lightweight foreground refinement method, and Canny edge detection is used to create a boundary-based evaluation signal.

3. Methodology

The final pipeline follows this modular sequence: input image, YOLO person detection, SAM box-prompted mask generation, Canny edge map, edge refinement, GrabCut refinement, and final edge-alignment scoring. The design is modular so that each experiment version can test a different combination of localization, segmentation, and refinement modules.

Photo Progress for the Final Experiment

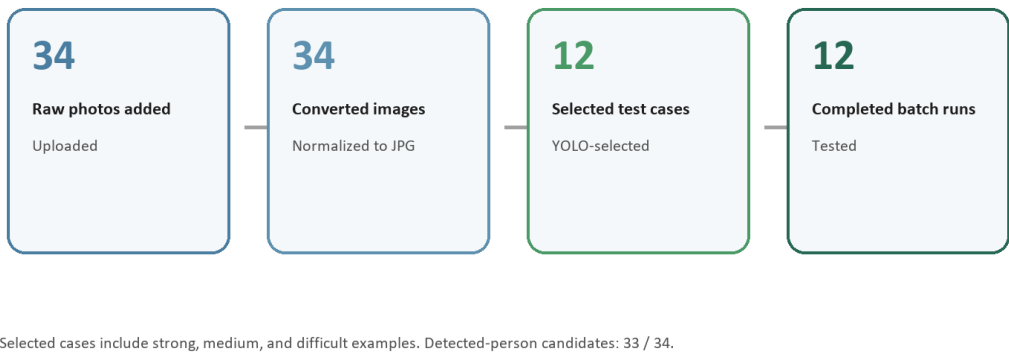


Figure 1. Dataset preparation and final batch testing progress.

4. Implementation

The final code package contains runnable scripts, model weights, selected test images, and complete batch outputs. The main implementation files are listed below.

File	Role
sam_boundary_refinement.py	Main SAM mask generation, Canny edge scoring, edge refinement, and GrabCut refinement.
run_experiment_versions.py	Runs v1-v8 model combinations for one image.
run_batch_test_images.py	Runs the experiment versions across all selected test images.
prepare_test_images.py	Converts and prepares uploaded test images.
requirements.txt	Lists the Python dependencies.

The model files used by the project are sam_vit_b_01ec64.pth, yolov8n.pt, and yolov8n-seg.pt. The final package stores these files under the model_weights folder so the project can be rerun from the final deliverable folder.

5. Experiment Setup

The experiment used 12 selected images from 34 uploaded photos. Each selected image was processed through the same batch testing script. The following table summarizes the tested versions.

Version	Modules	Runs	Mean	Role
v1	SAM auto + Canny + GrabCut	12	0.8708	Baseline
v2	SAM auto + Edge refine + GrabCut	12	0.8666	Baseline variant
v3	Manual box + SAM + GrabCut	0	-	Skipped

Version	Modules	Runs	Mean	Role
v4	Manual box + SAM + Edge refine	0	-	Skipped
v5	YOLO box + SAM + GrabCut	12	0.8422	YOLO-guided
v6	YOLO box + SAM + Edge refine + GrabCut	12	0.8855	Best current
v7	YOLO-Seg direct + Edge refine	12	0.5768	Weak direct mask
v8	YOLO-Seg + GrabCut	12	0.7918	Post-processing

6. Results

The best current version is v6, which combines YOLO detection, SAM, Canny edge scoring, edge refinement, and GrabCut. It achieved a mean edge alignment score of 0.8855. The minimum score of v6 is 0.7518, suggesting better stability than v5, whose minimum score dropped to 0.2756 in one failure case.

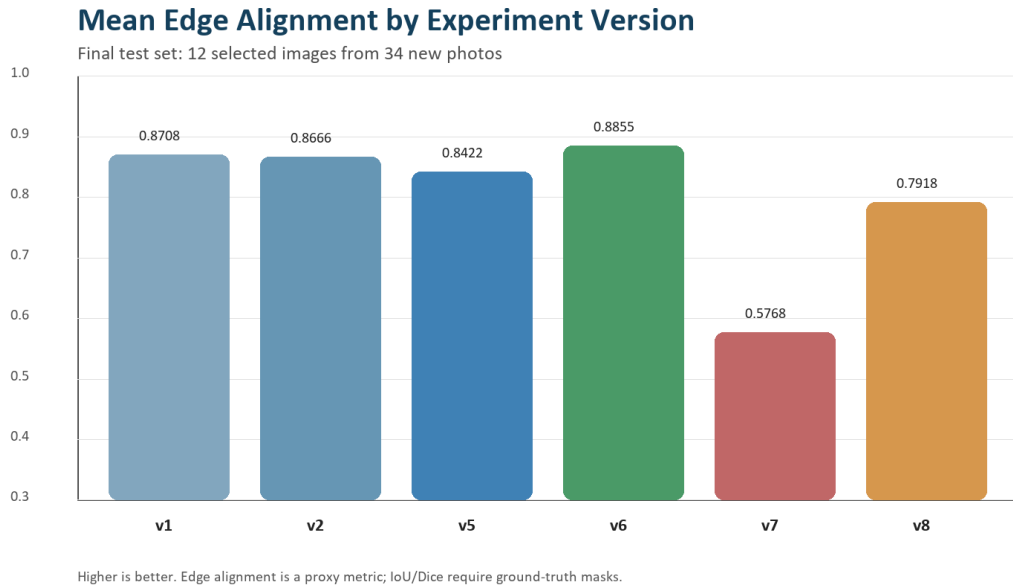


Figure 2. Mean edge alignment by experiment version.

YOLO-Seg alone performed the weakest, with a mean edge alignment score of 0.5768. Adding GrabCut to YOLO-Seg improved the score to 0.7918, showing that post-processing can improve direct segmentation masks. However, the YOLO-guided SAM pipeline still produced the strongest result in this experiment.

Representative Case: YOLO-Guided SAM Improves Boundary Alignment



Figure 3. Representative visual comparison of v1, v6, and v8.

7. Discussion

The experiment supports the idea that a hybrid model pipeline can be effective for small project segmentation tasks. YOLO performs person localization, SAM performs promptable mask generation, Canny edge detection provides a boundary-based evaluation signal, and GrabCut performs foreground-background refinement.

The main limitation is that the current evaluation does not include manually annotated ground-truth masks. Therefore, edge alignment should not be treated as final segmentation accuracy. The result is best described as a preliminary boundary-quality comparison.

Limitation Case: Some YOLO-Guided Variants Still Need Robustness



Figure 4. Limitation case showing that YOLO-guided variants still need robustness.

8. Conclusion

This project developed and tested a practical multi-model boundary refinement pipeline for person segmentation. Among the tested versions, YOLO-guided SAM with edge refinement and GrabCut achieved the highest mean edge alignment score on the selected 12-image test set. The result indicates that using YOLO as a localization step can reduce target ambiguity before SAM generates the mask, and lightweight post-processing can further improve boundary quality.

Future work should include manually labeled ground-truth masks, IoU and Dice evaluation, a larger test set, and comparison with stronger segmentation baselines.

References

- [1] A. Kirillov et al., "Segment Anything," Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023.
- [2] J. Terven and D. Cordova-Esparza, "A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS," Machine Learning and Knowledge Extraction, 2023.
- [3] Y. F. A. Gaus, N. Bhowmik, B. K. S. Isaac-Medina, and T. P. Breckon, "Performance Evaluation of Segment Anything Model with Variational Prompting for Application to Non-Visible Spectrum Imagery," arXiv:2404.12285, 2024.
- [4] M. A. Mazurowski et al., "Segment Anything Model for Medical Image Analysis: An Experimental Study," Medical Image Analysis, 2023.
- [5] C. Rother, V. Kolmogorov, and A. Blake, "GrabCut: Interactive Foreground Extraction Using Iterated Graph Cuts," ACM SIGGRAPH, 2004.
- [6] J. Canny, "A Computational Approach to Edge Detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, 1986.